Supporting Information

hTAS2R38 polymorphisms modulate oral microbiota and influence the prevalence and treatment outcome of halitosis

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Supplemental Figures

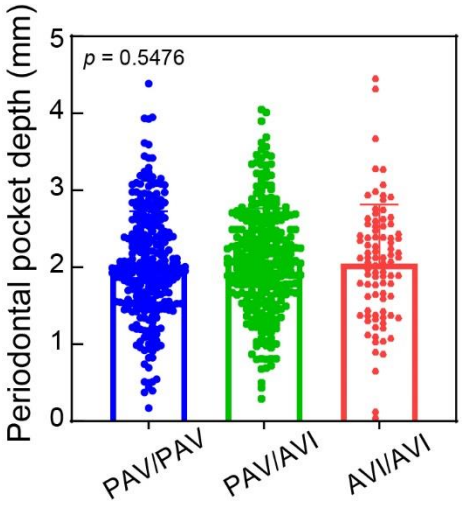


Figure S1. Periodontal pocket depth of individuals with different *hTAS2R38* genotype in the cross-sectional study. (Data are presented as mean and single points. Statistical comparisons were performed using one-way ANOVA.)

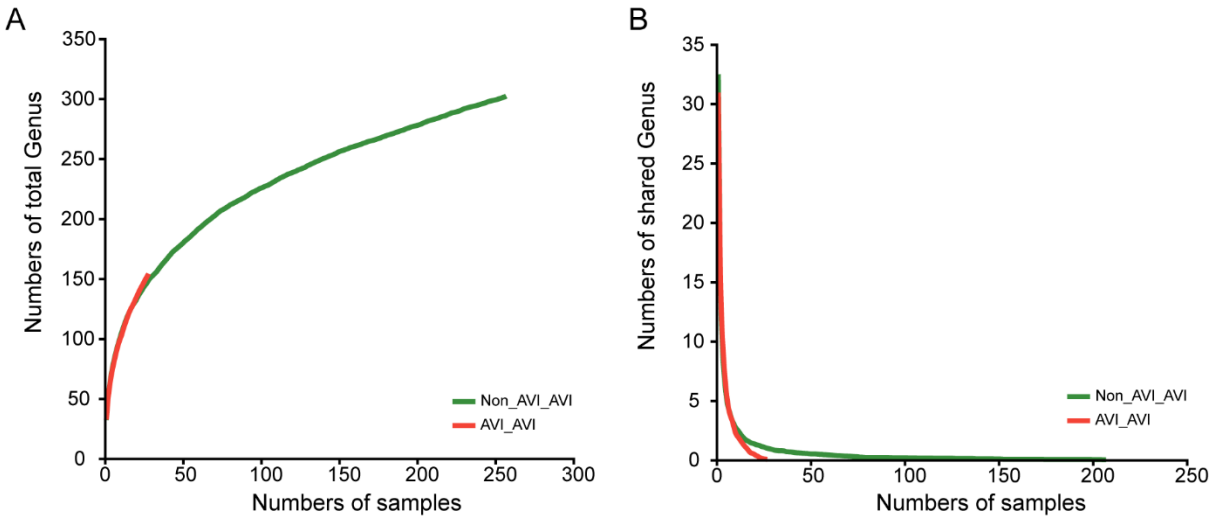


Figure S2. Sequencing sample assessment for the cross-sectional study. (A) Pan analysis. (B) Core analysis.

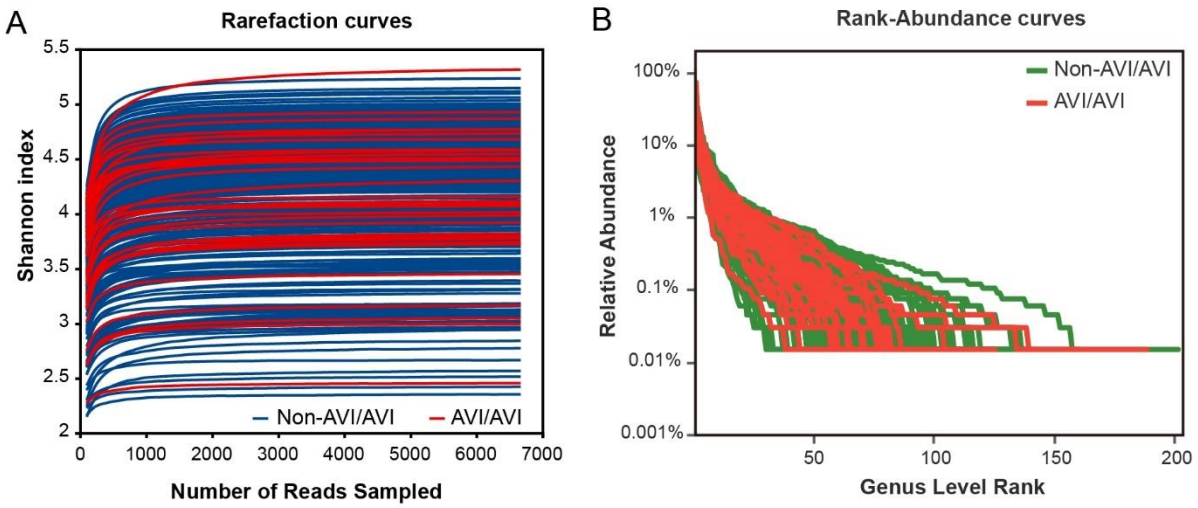


Figure S3. The 16S rRNA sequencing depth assessment in the cross-sectional study. A. Rarefaction Curves. B. Rank-Abundance Curves.

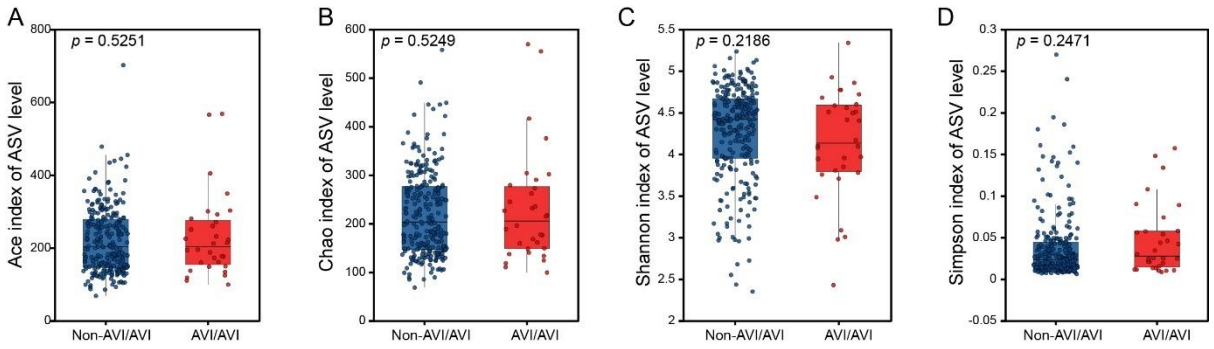


Figure S4. The α diversity of non-AVI/AVI and AVI/AVI groups in the cross-sectional study. Ace(A), chao(B), simpson(C), and shannon(D) index of ASV level in non-AVI/AVI and AVI/AVI groups. (Data are presented as mean ± SD. Statistical comparisons were performed using two-tailed Student's *t*-test.).

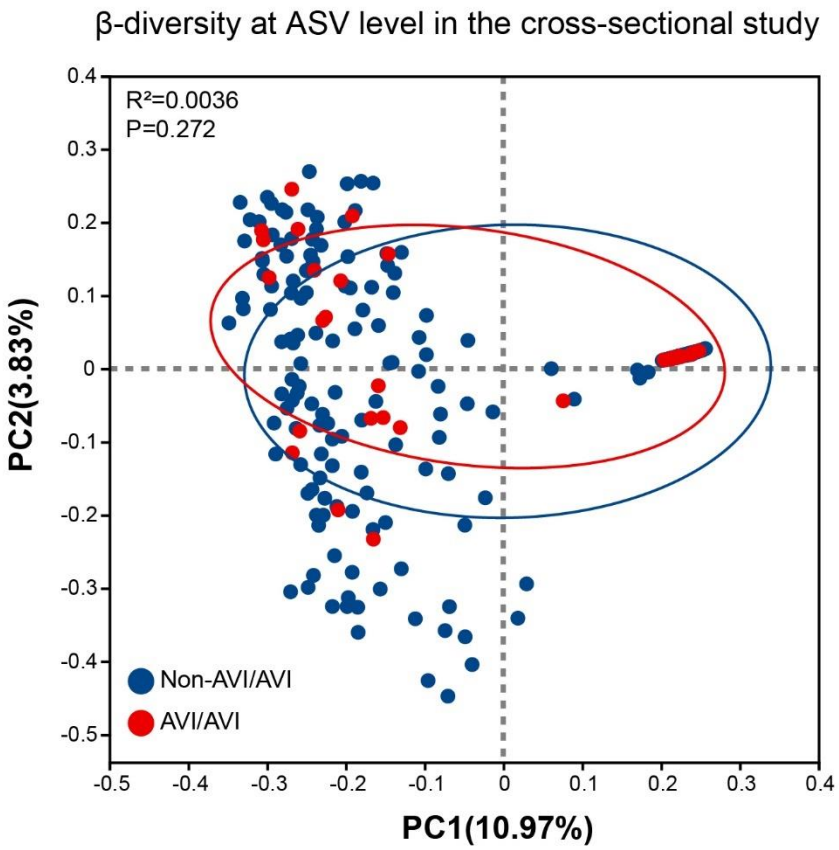


Figure S5. The principal coordinate (PCoA) based β diversity of non-AVI/AVI and AVI/AVI groups in the cross-sectional study. (Distance analysis used bray_curtis algorithm. Statistical analysis used adnois test, $P = 0.272$, $R^2 = 0.0036$).

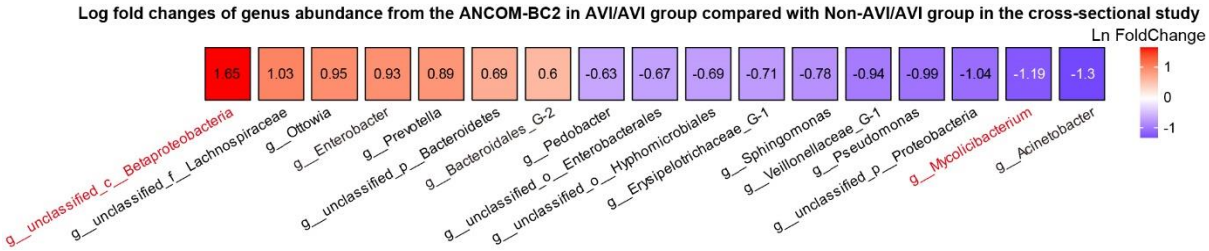


Figure S6. Log fold changes of genus with differential abundance screened by ANCOMBC2 in AVI/AVI group compared with Non-AVI/AVI group in the cross-sectional study. ($P<0.05$. Genus labeled in red indicates $P_{adj}<0.05$ as calculated by the Benjamini-Hochberg FDR method).

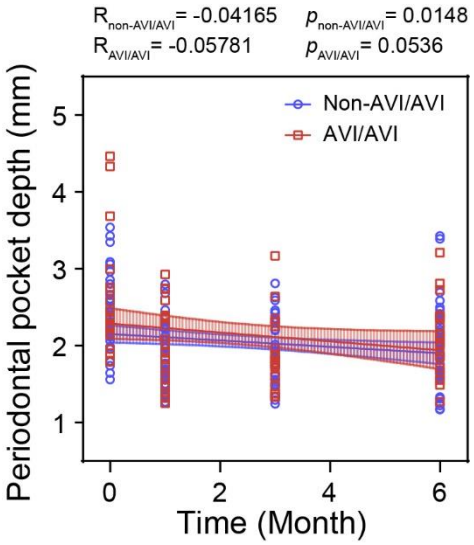


Figure S7. Periodontal probing depth (PPD) among individuals with different genotypes (non-AVI/AVI, and AVI/AVI) before and after periodontal treatment. (Linear regression analysis was used to analyze the changes in PPD after halitosis intervention. The lines represent the fitted linear regression lines, with shaded areas indicating the 95% CI. Red denotes the AVI/AVI group, while blue denotes the non-AVI/AVI group.)

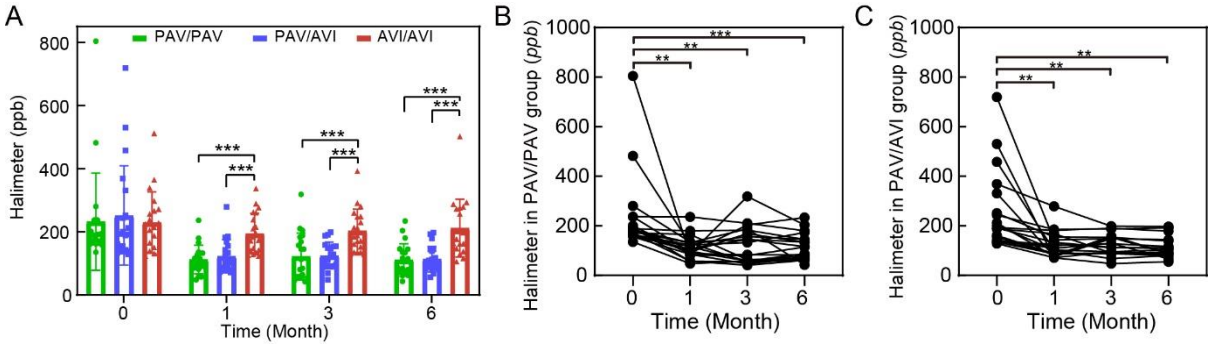


Figure S8. Halimeter scores among individuals with different genotypes (PAV/PAV, PAV/AVI, and AVI/AVI) before and after halitosis intervention. (A) Comparation of the halimeter scores of different genotypes at the same time points. Data were presented as mean ± SD with individual symbols. Statistical analysis was performed using unpaired student's *t* test. *, $P < 0.05$. **, $P < 0.01$. ***, $P < 0.001$. (B-C) The changes in halimeter scores in individuals with PAV/PAV (B) and PAV/AVI (C) genotypes at 1, 3, and 6 months after treatment. The data are represented as individual symbols, and statistical analysis was conducted using paired student's *t* test.

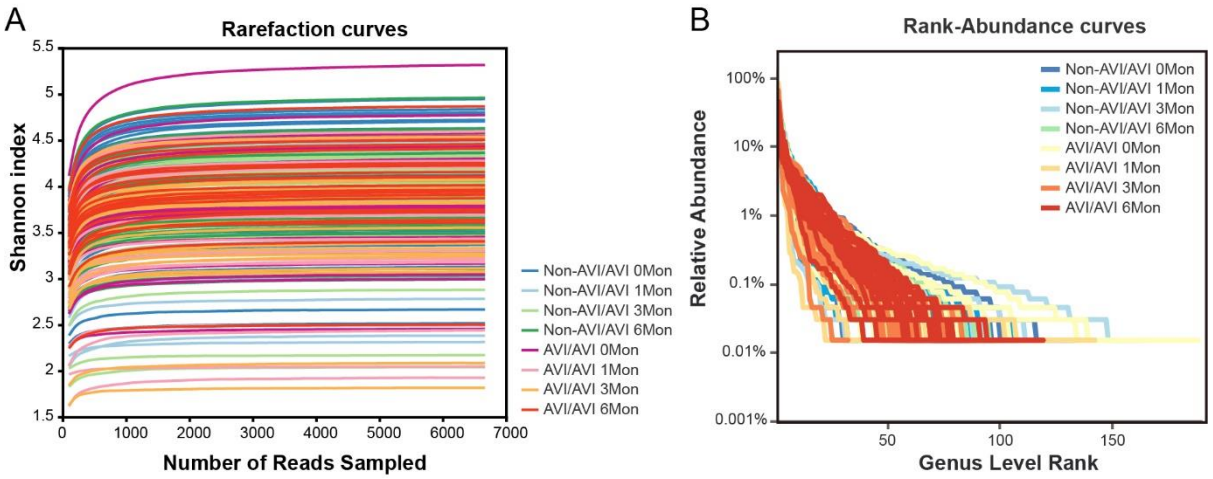


Figure S9. The 16S rRNA sequencing depth assessment in the prospective study. A. Rarefaction Curves. B. Rank-Abundance Curves.

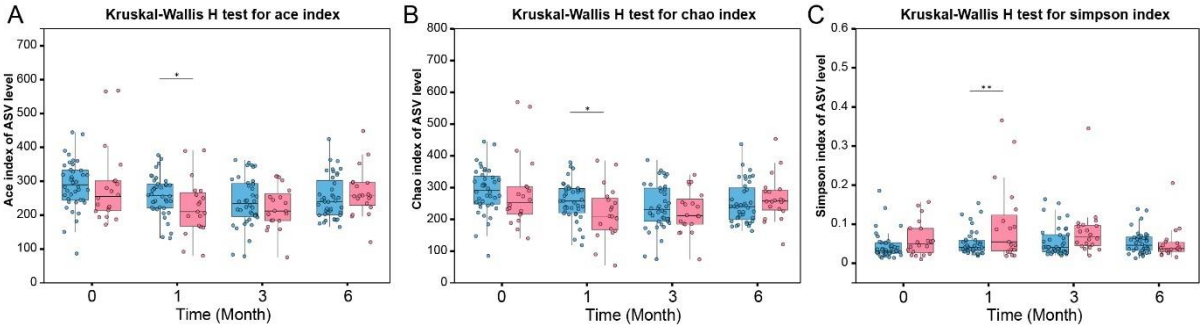


Figure S10. The α diversity of non-AVI/AVI and AVI/AVI groups before and after halitosis intervention. (A-C) α diversity index of Ace(A), Chao(B), and Simpson(C) at ASV level in non-AVI/AVI and AVI/AVI groups. Data were presented as mean ± SD. Statistical analysis was performed using unpaired student's *t* test. *, $P < 0.05$.

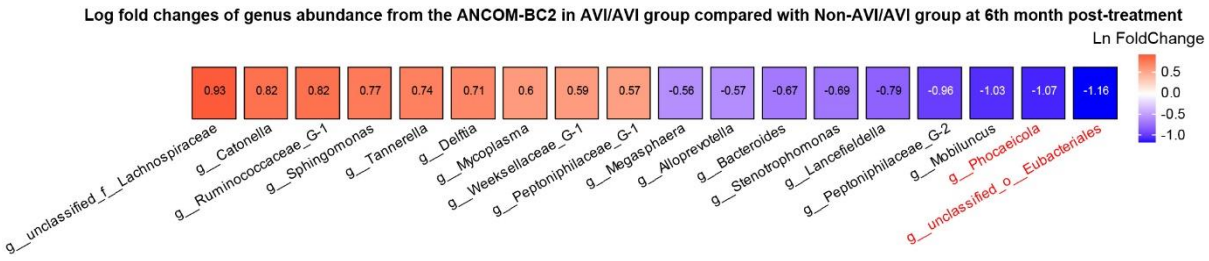


Figure S11. Log fold changes of genus with differential abundance screened by ANCOMBC2 in AVI/AVI group compared with Non-AVI/AVI group in the prospective study ($P<0.05$. Genus labeled in red indicates $P_{adj}<0.05$ as calculated by the Benjamini-Hochberg FDR method).

Tables

Table S1. Distribution of volatile sulfur compounds (VSCs) level compared with organoleptic judgment by a trained professional at different cut-off values (n = 20).

		organoleptic judgment	
		Halitosis	Healthy
	Halitosis	8	5
Halimeter test	Healthy	0	7
	Total	8	12
Cut off = 100 *ppb*	**Kappa**	**0.53**	
	Halitosis	7	2
Halimeter test	Healthy	1	10
	Total	8	12
Cut off = 120 *ppb*	**Kappa**	**0.69**	
	Halitosis	4	1
Halimeter test	Healthy	4	11
	Total	8	12
Cut off = 140 *ppb*	**Kappa**	**0.44**	

Table S2. Sample size calculation for the cross-sectional study (n = 40).

	Healthy (21)	Halitosis (19)
PAV/PAV	8	7
PAV/AVI	11	9
AVI/AVI	2	3

Table S3. Sample size calculation for the prospective study (n = 15).

		Non-AVI/AVI (n = 10)		AVI/AVI (n = 5)
		PAV/PAV	PAV/AVI	
Treatment outcome	**Improved**	3	3	1
	Persistent	2	2	4

Table S4. Primers for *hTAS2R38*

Primer	Sense	Antisense
rs713598	CTAACTCGCATCCGCACTG	GCAGCAAGCCAGAGGTTG
rs1726866+rs10246939	GGCACATGAGGACAATGAAGG	GTCCATTCTCAGCACAGTGTC

Table S5. Diagnostic distribution of halitosis by different examiners using Halimeter®.

		Examiner 1	
		Halitosis	**Non-halitosis**
Examiner 2	**Halitosis**	8	1
	Non-halitosis	0	11
Total		8	12
Kappa		**0.90**	

Table S6. Specific primer sequences of *Prevotella intermedia*

Primer	**Sense**	**Antisense**
Universal	GAGCCGATAAGGACGTGGTA	TTTAGGCTTGACCCATTTCG
Prevotella intermedia	TTTGTTGGGGAGTAAAGCGGG	TCAACATCTCTGTATCCTGCGT

Table S7. Multiple comparisons of genotypes distribution in cross-sectional study

Pairwise comparison	*P* value *	OR#	95% Confidence Interval	
			Lower	Upper
AVI/AVI-PAV/AVI	0.0020	2.167	1.324	3.497
AVI/AVI-PAV/PAV	0.0001	2.622	1.597	4.244
PAV/AVI-PAV/PAV	0.2448	1.210	0.8772	1.673

* Adjusted significance level $\alpha' \leq 0.017$.

OR: odds ratio

Table S8. Association between *hTAS2R38* Genotype and Halitosis

Halitosis Case classification	Gene Distribution		*P* value*	OR#
	AVI/AVI	non-AVI/AVI (PAV/AVI+PAV/PAV)		
Halitosis (n=339)	60	279 (148+131)	0.0002	2.380
Non halitosis (n=350)	29	321 (155+166)		

* *P* value for 2×2 contingency table, using Chi-square test.

OR: odds ratio

Table S9. Clinical characteristics between the population receiving 16S rRNA sequencing (n = 290) and the general population (n = 689) in the cross-sectional study.

		Sequencing population	General population	*P* value
Age		32.75 ± 6.69	32.49 ± 7.88	0.6197 [a]
Gender	**Male**	148	321	0.2036 [b]
	Female	142	368	
Halimeter Scores		148.85 ± 144.90	143.86 ± 116.06	0.5701 [a]
PPD (mm)		2.08 ± 0.68	2.06 ± 0.69	0.7231 [a]
Genotype	**PAV/PAV**	130	297	0.6958 [b]
	PAV/AVI	128	303	
	AVI/AVI	32	89	

[a] Two sample unpaired student's *t* test. [b] Chi-square test.

Table S10. Gene Distribution and Hardy-Weinberg Equilibrium Test for individuals performing 16S rRNA sequencing in the cross-sectional study.

Sample	Gene Distribution			Allele distribution		*P* value*
	AVI/AVI	PAV/AVI	PAV/PAV	AVI	PAV	
290	32	128	130	192	388	0.6679

* *P* value for 2×3 contingency table, using Chi-square test

Table S11. Up-regulated and down-regulated oral microbes at genus level in AVI/AVI group compared with non-AVI/AVI group by DESeq2 analysis. (Screening parameters: $P_{adj} < 0.05$)

Genus	Base Mean	log2Fold Change	*P* value	P_{adj}[a]	Regulated
Prevotella	437.7964	1.522918	3.57E-05	0.002663	Up
Agrobacterium	8.316963	-3.79974	3.41E-05	0.002663	Down
Leptothrix	16.88989	-2.78124	9.63E-05	0.004602	Down
Oryzomicrobium	11.64956	-3.89634	0.000124	0.004602	Down
Brucella	6.91172	-4.50329	0.000362	0.010793	Down
unclassified_f__Bradyrhizobiaceae	71.15244	-3.86498	0.000466	0.011563	Down
Pseudomonas	49.82809	-2.55832	0.000597	0.012712	Down
Brevundimonas	23.59856	-1.93638	0.002038	0.037964	Down

a. P_{adj} was calculated by the Benjamini-Hochberg FDR method

Table S12. Genus with differential abundance screened by ANCOMBC2 in AVI/AVI group compared with non-AVI/AVI group in the cross-sectional study. (Screening parameters: $P < 0.05$)

Taxon	Ln Fold Change (lfc)	W (lfc/se)	*P* value	P_{adj} [a]
g__unclassified_c__Betaproteobacteria	1.65	5.369347	2.52E-05	0.003451
g__Mycolicibacterium	-1.19	-3.62995	0.000647	0.044353
g__Acinetobacter	-1.3	-3.16047	0.002114	0.072391
g__unclassified_p__Proteobacteria	-1.04	-3.25433	0.00183	0.072391
g__Enterobacter	0.93	3.057183	0.004766	0.108812
g__unclassified_f__Lachnospiraceae	1.03	2.972925	0.003994	0.108812
g__Ottowia	0.95	2.604564	0.011858	0.20548
g__Pseudomonas	-0.99	-2.56194	0.011999	0.20548
g__Veillonellaceae_G-1	-0.94	-2.46174	0.015033	0.228835
g__unclassified_o__Hyphomicrobiales	-0.69	-2.4264	0.018684	0.255974
g__unclassified_o__Enterobacterales	-0.67	-2.34713	0.022839	0.284449
g__Sphingomonas	-0.78	-2.24434	0.02772	0.316473
g__Bacteroidales_G-2	0.6	2.170113	0.032662	0.319618
g__Prevotella	0.89	2.179877	0.030837	0.319618
g__Erysipelotrichaceae_G-1	-0.71	-2.08398	0.039544	0.338594
g__Pedobacter	-0.63	-2.1245	0.037902	0.338594
g__unclassified_p__Bacteroidetes	0.69	2.072892	0.043356	0.3494

a. P_{adj} was calculated by the Benjamini-Hochberg FDR method

Table S13. Genotype distribution and treatment outcome of halitosis patients with different genotypes in the prospective study.

Characteristics		Genotype			*P* value
		PAV/PAV (n=20)	PAV/AVI (n=20)	AVI/AVI (n=20)	
Treatment outcome	Improved	13	14	2	0.0001 [a]
	Persistent	7	6	18	

[a] 2×3 Chi-square test.

Table S14. Clinical characteristics and treatment outcomes of halitosis patients with non-AVI/AVI genotypes after halitosis interventions in the prospective study.

Characteristics		Genotype		*P* value
		PAV/PAV (n=20)	PAV/AVI (n=20)	
Age, mean±SD, y		34.35 ± 6.64	33.30 ± 6.61	0.6279 [a]
Male sex, n (%)		6 (30.0)	10 (50.0)	0.1967 [b]
Halimeter (*ppb*)	Baseline	231.90 ± 149.79	251.32 ± 158.43	0.6951 [a]
	1-month	111.58 ± 43.85	122.43 ± 50.52	0.4839 [a]
	3-month	121.48 ± 72.71	124.43 ± 40.99	0.8784 [a]
	6-month	110.68 ± 50.23	112.68 ± 39.67	0.8924 [a]

[a] Student *t*' test. [b] Chi-square test.

Table S15. Up-regulated and down-regulated oral microbes at the genus level in the AVI/AVI group compared with the non-AVI/AVI group at the 6th month after treatment. (Screening parameters: $P_{adj} < 0.05$)

Genus	BaseMean	Log2Fold Change	*P* value	P_{adj} [a]	Regulated
Filifactor	25.6665	2.542286	2.89E-05	0.004892	up

a. P_{adj} was calculated by the Benjamini-Hochberg FDR method

Table S16. Genus with differential abundance screened by ANCOMBC2 in AVI/AVI group compared with Non-AVI/AVI group at the 6th month after treatment. (Screening parameters: $P < 0.05$)

Taxon	Ln Fold Change (lfc)	W (lfc/se)	*P* value	P_{adj} [a]
g__unclassified_f__Lachnospiraceae	0.93	3.616876	0.00172	0.064771
g__Delftia	0.71	3.123	0.005358	0.109105
g__Catonella	0.82	2.633498	0.011957	0.16889
g__Mycoplasma	0.6	2.218215	0.038923	0.282889
g__Sphingomonas	0.77	2.193789	0.037387	0.282889
g__Tannerella	0.74	2.185889	0.033951	0.282889
g__Weeksellaceae_G-1	0.59	2.215784	0.036039	0.282889
g__Ruminococcaceae_G-1	0.82	2.073295	0.043116	0.286595
g__Peptoniphilaceae_G-1	0.57	2.11589	0.048553	0.304721
g__unclassified_o__Eubacteriales	-1.16	-4.95	4.24E-05	0.004793
g__Phocaeicola	-1.07	-3.89604	0.000833	0.047051
g__Mobiluncus	-1.03	-3.01722	0.005793	0.109105
g__Stenotrophomonas	-0.69	-3.09702	0.010158	0.16398
g__Peptoniphilaceae_G-2	-0.96	-4.18591	0.002355	0.066539
g__Bacteroides	-0.67	-2.69089	0.016069	0.201753
g__Megasphaera	-0.56	-2.44095	0.02461	0.278097
g__Alloprevotella	-0.57	-2.13753	0.040055	0.282889
g__Lancefieldella	-0.79	-2.203	0.031726	0.282889

a. P_{adj} was calculated by the Benjamini-Hochberg FDR method

Table S17. The MaAsLin-based correlations between the relative abundance of genus and time throughout the treatment ~~regimen~~ in Non-AVI/AVI group. (Screening parameters: $P < 0.05$ and $P_{adj} < 0.2$)

Feature	Coefficient	N	N not 0	*P* value	P_{adj} [a]
Mycoplasma	-0.00165	160	59	0.001597	0.060662
Phocaeicola	-0.00161	160	70	0.002091	0.060662
Fretibacterium	-0.00362	160	104	0.00292	0.060662
unclassified_k__Bacteria	-0.0017	160	75	0.004333	0.060662
Filifactor	-0.00339	160	135	0.007451	0.082283
Peptostreptococcaceae_G-6	-0.00157	160	100	0.008235	0.082283
Treponema	-0.00283	160	94	0.008396	0.082283
Tannerella	-0.00213	160	122	0.010745	0.087124
Peptostreptococcaceae_G-4	-0.00273	160	119	0.011754	0.087124
Alloprevotella	-0.00192	160	93	0.012775	0.087124
Veillonellaceae_G-1	-0.00396	160	150	0.016821	0.097562
unclassified_f__Peptostreptococcaceae	-0.00088	160	53	0.023596	0.121703
unclassified_f__Lachnospiraceae	-0.00119	160	64	0.030959	0.145753
Lachnospiraceae_G-8	-0.00141	160	104	0.031233	0.145753
Actinomyces	0.004679	160	156	0.003114	0.060662
Streptococcus	0.013448	160	160	0.003356	0.060662
Schaalia	0.012252	160	158	0.003871	0.060662
Solobacterium	0.003182	160	148	0.011905	0.087124
Lancefieldella	0.002878	160	146	0.013335	0.087124
Mogibacterium	0.001429	160	136	0.016924	0.097562
Peptostreptococcaceae_G-1	0.002703	160	146	0.019813	0.107872

a. P_{adj} was calculated by the Benjamini-Hochberg FDR method